

Roll No.

TEE-701

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

ELECTRICAL DRIVES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) State and explain the function of various converters used in electric drives. (CO1, CO2)
- (b) What are the main factors which decide the choice of electrical drive for a given application ? (CO1, CO2)
- (c) Discuss the thermal model of motor for heating and cooling with suitable diagram. (CO1, CO2)
2. (a) Draw and explain four quadrant operation of a drive system using suitable example. (CO3, CO4)

P. T. O.

- (b) A motor operates on a periodic duty cycle in which it is clutched to its load for 10 min and declutched to run on no-load for 20 min. Minimum temperature rise is 40°C . Heating and cooling time constants are equal and have a value of 60 min. When load is declutched continuously the temperature rise is 15°C .

Determine :

(CO3, CO4)

- (i) Maximum temperature during the duty cycle, and
 - (ii) Temperature when the load is clutched continuously.
- (c) Determine the fundamental torque equation used for motor load system.

(CO3, CO4)

3. (a) Explain the operation of closed speed control scheme with inner current control loop. What are various function of inner current control loop ?

(CO5, CO1)

- (b) With the help of suitable block diagram explain V/f control used in induction motor drives incorporating closed loop system. (CO5, CO1)
- (c) A motor drives two loads. One has rotational motion. It is coupled to the motor through a reduction gear with $a = 0.1$ and efficiency of 90%. The load has a moment of inertia of 10 kg-m^2 and a torque of 10 N-m . Other load has translational motion and consists of 1000 kg weight to be lifted up at an uniform speed of 1.5 m/s. Coupling between this load and the motor has an efficiency of 85%. Motor has an inertia of 0.2 kg-m^2 and runs at a constant speed of 1420 rpm. Determine equivalent inertia referred to the motor shaft and power developed by the motor.

(CO5, CO1)

(3)

4. (a) Discuss regenerative and dynamic braking. What are drawback of dynamic braking ? (CO3, CO6)
- (b) Discuss the constant torque drive and constant power drive. (CO3, CO6)
- (c) A drive has the following parameters : (CO3, CO6)

$J = 10 \text{ kg-m}^2$, $T = 100 - 0.1N$, N-m Passive load torque

$T_f = 0.05N$, N-m, where N is the speed in rpm.

Initially the drive is operating in steady-state. Now it is to be reversed. For this motor characteristic is changed to $T = -100 - 0.1N$, N-m. Calculate the time of reversal.

5. (a) Discuss various types of load characteristics in details. (CO2)
- (b) A 200V, 10.5A, 2000 rpm shunt motor has the armature and field resistances of 0.5 and 400Ω respectively. It drives a load whose torque is constant at rated motor torque. Calculate motor speed if the source voltage drops to 175 V. (CO2)
- (c) Discuss the characteristics of various types of dc motor used in drives. (CO2)

Roll No.

TEE-702

**B. TECH. (EE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

**NON-CONVENTIONAL RESOURCES AND
ENERGY SYSTEM (NCR AND ES)**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the concept of energy and how it is related to economy and social development. (CO1 & CO5)
(b) Explain the concept of Renewable energy resources. (CO1 & CO5)
(c) Give ad detail note on fuel cell. (CO1 & CO5)
2. (a) What are the applications of geothermal energy ? (CO4, CO2 & CO1)
(b) Explain the types of solar collectors. (CO4, CO2 & CO1)
(c) What are the environmental aspects of energy ? (CO4, CO2 & CO1)

P. T. O.

(2)

3. (a) What are the major applications of wind power ? (CO3 & CO6)
- (b) What do you mean by Magneto-hydrodynamic power conversion ?
(CO3 & CO6)
- (c) What are the usable forms of bio mass ? (CO3 & CO6)
4. (a) Explain the types of wind turbine. (CO3, CO2 & CO6)
- (b) What is solar PV system ? (CO3, CO2 & CO6)
- (c) What is Thermoelectric power conversion ? (CO3, CO2 & CO6)
5. (a) Explain Solar Thermal System and its application. (CO2, CO4 & CO1)
- (b) What are the types of geothermal resources ? (CO2, CO4 & CO1)
- (c) Compare between Conventional and Renewable energy resources.
(CO2, CO4 & CO1)

Roll No.

TOE-703(A)

**B. TECH. (EE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

BIOMEDICAL ELECTRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Differentiate action potential and resting potential. Sketch the action potential neat and clean waveform. (CO1)
- (b) What are the components of a biomedical system ? Explain any *three*. (CO1)
- (c) Summarize active and passive transducers. Generalize the property of piezoelectric transducer. (CO2)
2. (a) Classify the various types of electrodes. Point out any *four* types of surface electrodes. Draw the electrode configuration of AVR output. Name the electrodes used for EEG measurement. (CO1)

P. T. O.

(2)

- (b) Discuss briefly about EMG. What is evoked response ? (CO2)
- (c) Summarize and define the Einthoven Triangle. (CO2)
- 3. (a) Sketch the block diagram of ventilators SYSTEM and explain the functions of each block. (CO3)
- (b) What are the methods for measuring blood pressure ? Sketch a typical setup and explain briefly. (CO3)
- (c) Explain the structure of physiology of respiratory system. (CO3)
- 4. (a) Explain the measurement method of ESR with a neat diagram. (CO4)
- (b) Discuss briefly about Tomography. What is computer aided tomography ? (CO4)
- (c) How X-ray are produced ? What is its properties ? (CO4)
- 5. (a) Summarize the instruments used to check the safety parameters of biomedical equipment. (CO1)
- (b) With neat block diagram explain the principle of operation and working of MRI system in medical terms. (CO4)
- (c) Prepare the procedure of hemodialysis with suitable block diagram. (CO4)

Roll No.

TOE-704

**B. TECH. (EE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

FUNDAMENTALS OF IoT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain how power consumption and battery life is a major challenge in a IoT based environmental monitoring system, discuss the impact and potential mitigation approach. (CO1 & CO2)
- (b) Compare Automation and Internet of things using table with suitable example. (CO1 & CO2)
- (c) Explain following terminologies : (CO1 & CO2)
 - (i) Smart device
 - (ii) Encryption

P. T. O.

2. (a) Consider yourself an IoT architect, you have to design a schematic layout with respect to various layers in IoT system for a smart agriculture farming application. (CO2 & CO3)
- (b) Explain working definition of IoT framework and discuss any *two* basic nodal capabilities of it. (CO2 & CO3)
- (c) Explain following terminologies : (CO2 & CO3)
- (i) User Interface
 - (ii) Microcontroller unit
3. (a) Explain various methods of identification of IoT objects. (CO3 & CO5)
- (b) Compare star and mesh topologies of IoT connections. Which topology will be more suitable for a smart traffic management IoT systems ? (CO3 & CO5)
- (c) What do you mean by data localization and traffic pattern in traffic characteristics of IoT system ? (CO3 & CO5)
4. (a) How Internet Protocol (IP) plays crucial role in IoT systems ? Explain any three key uses of IP in IoT systems. (CO4)
- (b) What is Wireless Sensor Network (WSN) ? Explain its architecture using suitable diagram. (CO4)
- (c) Explain any *three* applications of wireless sensor networks use suitable diagram to elaborate the applications. (CO4)

(3)

5. (a) What are the software agents ? What is their purpose in Internet of Things based applications ? (CO2 & CO6)
- (b) What do you mean by identity portrayal in IoT ? Describe various identity management models used in IoT based systems. (CO2 & CO6)
- (c) Design a model application for an IoT enabled smart bottle refilling and packaging system for beverage company. Define the resources, features, and block diagram of the proposed system. (CO2 & CO6)

Roll No.

TEE-706

**B. TECH. (EE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

UTILIZATION OF ELECTRICAL ENERGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain in brief how heating is done in the following cases : (CO1)
- (i) Resistance heating
 - (ii) Induction heating
- (b) Write a short note on High Frequency Eddy-current Heating. (CO1)
- (c) A 15-kW, 220-V, single-phase resistance oven employs nickel-chrome wire for its heating elements. If the wire temperature is not to exceed $1,000^{\circ}\text{C}$ and the temperature of the charge is to be 600°C , calculate the diameter and length of the wire. Assume radiating efficiency to be 0.6 and emissivity as 0.9. For nickel chrome resistivity is $1.016 \times 10^{-6} \Omega\text{-m}$. (CO1)

P. T. O.

2. (a) Explain the various types of lighting schemes with relevant diagrams. (CO2)
- (b) Estimate the number and wattage of lamps which would be required to illuminate a workshop space 60×15 metres by means of lamps mounted 5 metres above the working plane. The average illumination required is about 100 lux. Coefficient of utilization = 0.4; Luminous efficiency = 16 lm/W. (CO2)
- Assume a spacing/height ratio of unity and a candle power depreciation of 1/1.2
- (c) Enumerate the various factors, which have to be considered while designing any lighting scheme. (CO2)
3. (a) Discuss the formation and characteristics of Electric Arc in Welding Process. What is the effect of arc length ? (CO3)
- (b) Discuss the need of arc welding machines. (CO3)
- (c) Compare Resistance Welding and Arc Welding. (CO3)
4. (a) What are the various advantages and disadvantages of Electric Traction ? (CO4)
- (b) Explain the requirements for ideal traction and show which drive satisfies almost all requirements. (CO4)
- (c) What are the various systems of Railway Electrification ? (CO4)

(3)

5. (a) Discuss general features of Traction Motor. A train runs at an average speed of 45 km per hour between stations 2.5 km apart. The train accelerates at 2 km/h/s and retards at 3 km/h/s. Find its maximum speed assuming a trapezoidal speed/time curve. Calculate also the distance travelled by it before the brakes are applied. (CO5)
- (b) What are various types of traction motors ? (CO5)
- (c) Explain the terms tractive effort, coefficient of adhesion, train resistance and specific energy consumption of train. (CO5)

